Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	**HTML, CSS, JavaScript, Bootstrap**	Creates a responsive, user-friendly interface for entering soil and environmental data and viewing crop recommendations.
2	**Backend / Server-Side**	**Python, Flask**	Flask provides a lightweight web framework, while Python supports machine learning model integration and data processing.
3	**Database / Data Storage**	**MySQL**	Stores user information, crop datasets, soil parameters, and prediction history efficiently.
4	**Cloud / Hosting / Deployment**	**Render / PythonAnywhere / Heroku**	Enables easy deployment, remote access, and scalable hosting of the web application.
5	**Version Control & CI/CD**	**Git, GitHub**	Supports source code management, team collaboration, version tracking, and project maintenance.
6	**Third-Party APIs / Other Tools**	**Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn**	Scikit-learn builds the crop prediction model, Pandas and NumPy process data, while Matplotlib and Seaborn visualize agricultural insights.